

NITESH M

Sr Data Engineer – Real time Data & MLOps Solutions

E: NITESHHCSE77@GMAIL.COM

C: 937-343-4126

<https://www.linkedin.com/in/niteshcse77>

Professional Experience

Sr Data-MLOps Engineer with experience in **architecting scalable cloud-native data and ML platforms across AWS, Snowflake, Open table formats, Kafka** for Fortune 500 clients. Designed real-time and batch processing frameworks **handling 10TB+ workloads** and millions of daily events, **reducing pipeline latency by 80%**, improving **query performance by 5x**, and **lowering operational costs by \$150K+ annually**.

Experienced in building and operating ML-enabled systems using K8s, Docker, **Airflow, SageMaker**, and **PySpark**, delivering scalable Data/ML infrastructure with 99.99% uptime, **80% faster deployments**, and **65% fewer pipeline failures**. Proven ability to productionize machine learning workloads through automation, observability, governance, and distributed systems engineering, improving model accuracy by 35% and accelerating audit readiness by 60%.

Clients:

Hyatt, TD Bank, Accenture, FL – Dept. Financial Services, Ryder Logistics, Windhaven Insurance Company, 3M

Certifications

- Databricks Certified Data Engineer Professional • NITESH CHANDRA MADHAM • Databricks Badges
- Databricks AWS Platform Architect
- Python for Data Science - Credly
- Snowflake- Hands On Essentials - Data Engineering - Credly
- Python Programming - Udemy Course Completion Certificate | Udemy
- BUILD 2023 Snowflake Builder Badge • Nitesh Madham • Snowflake Developer Badges

SKILLS

- MLOps & ML Platforms: **MLflow, Feature Engineering**, Real-Time Inference, ML Observability
- Programming: **Python**, Advanced SQL, **PySpark**, API Development, **Snowpark**
- Data Platforms and Streaming: Databricks, **Snowflake, OpenSearch, Redshift**, Oracle, SQL Server, MYSQL, Netezza, MongoDB, **Postgres**.
- Open Table formats: **Iceberg, Delta**
- Cloud Platforms: **AWS (SageMaker, EMR, Lambda)**, Kafka, Kinesis, Azure
- ETL Tools: Informatica, Talend, SSIS, **Glue**, DMS
- ML Frameworks: Scikit-Learn, **XGBoost**
- Tools: Docker, **Kubernetes**, Git, **Jupyter Notebooks**, **MLflow, Airflow, Astronomer**.
- Methodologies: Agile, DevOps, **CI/CD**

WORK EXPERIENCE

Client: Hyatt 11/2024 to current

Role: Sr. Data Engineer V (Real-Time Compliance)

Project: Guest Experience Management

- Implemented end-to-end lineage tracking** using Snowflake Tasks, Dynamic tables, and built Tableau dashboards, reducing SLA breaches by 45% and enabling compliance teams to trace suspicious transactions across 7 Hyatt global regions in near real-time.
- Integrated Kafka, S3, Snowflake and Tableau for **real-time processing and archival of guest data** from various hotels across the globe, achieving 99.99% uptime and **reducing backend ETL latency by 80%**.
- Led migration** of complex transformation logic to Snow-SQL and SnowSpark, **converting 60+ legacy stored procedures to optimized DataFrame operations** with predicate pushdown, **achieving 3.5x performance improvement**.
- Automated deployment of compliance pipelines via Git, AWS Lambda, Kubernetes ensuring consistent governance controls across Hyatt’s enterprise systems.
- Implemented Gitlab-iaaC** for all data pipelines, **reducing deployment time by 80%** and ensuring consistent environments **across regions**.

Client: TD-Bank (Anti Money Laundering – MLOps) 07/2023 to 11/2024

Role: Sr. Data/ML Engineer L3 (Risk Monitoring)

Project: EDW & Risk Monitoring Platform

- Migrated on-premise data warehouse to Redshift**, improving query performance by 40% and reducing operational costs by \$150,000 annually.
- Architected real time AML data pipelines using Snowflake, Kafka, and Airflow, processing **15M+ daily banking transactions** and reducing suspicious activity detection latency by **78%** across fraud and compliance workflows.
- Developed feature engineering** and transaction enrichment pipelines using Snowflake, Snowpark, and AWS services **to generate 150+ risk indicators** (transaction velocity, geographic anomalies, behavioral patterns), **improving suspicious activity prediction accuracy by 32%**.
- Implemented Data governance principles** in Snowflake and row-level access policies in Tableau to protect PII/PCI and Risk Ranking data **reducing unauthorized access incidents by 65%**.
- Engineered scalable feature engineering and transaction enrichment pipelines using Snowflake, Kafka, and AWS services to support XGBoost-based AML anomaly detection models, **contributing to a 32% increase in detection accuracy and reducing manual review cycles by 45%**.
- Supported the operationalization of **XGBoost-driven AML anomaly detection models** by building **scalable transaction enrichment and feature engineering pipelines**, contributing to an **F1 score of 0.85** and improving early detection of suspicious account behavior.

Client: Accenture - Bridgestone

Role: Sr. Data Engineer IV, HyperCare Lead

08/2022 to 07/2023

Project: Data Architecture & AML, Suspicious Activity Monitoring

- **Migrated legacy ETL platform to AWS using Glue, Python, PySpark functions and loading into Snowflake, reducing data processing time by 6 hours and handling 1.2 million batched records daily.**
- **Led a cross-functional team in migrating legacy data warehouse to Snowflake, resulting in 40% cost reduction and 65% improvement in query performance while implementing comprehensive data governance frameworks.**
- **Architected a petabyte scale AWS lakehouse using Apache Iceberg, supporting 300+ concurrent analytical workloads and improving large-table query performance by 5x through partition evolution, metadata pruning, optimized sort strategies, and automated file compaction.**
- **Developed feature engineering and enrichment workflows using Snowflake, Snowpark, and vector-based similarity search, generating 100+ operational indicators supporting predictive maintenance and anomaly identification initiatives.**
- **Collaborated with ML and AI platform teams to operationalize LLM and embedding-based workflows using vector stores, MLflow, SageMaker, and Snowflake feature datasets, improving search relevance and reducing inference response time by 35%.**
- **Implemented a scalable machine learning pipeline using Docker, Kubernetes, and AWS EKS, enhancing model deployment efficiency by 40% and ensuring high availability of ML services.**
- **Contributed to the development of a customer churn prediction model using ensemble methods (Random Forest, XGBoost) in scikit-learn, helping clients reduce churn rates by an average of 15%.**

Client: FL - Department of Financial Services

01/2022 to 07/2022

Role: Sr. Data Engineer III & Data Lead

Project: Company Takeover & ETL process Modernization

- **Worked with terabytes of data from various insurance companies operated in multiple states which filed for bankruptcy taken over by FL-State.**
- **Led the migration of a 100TB risk management data warehouse from on-premises to Snowflake, optimizing table designs and implementing column-level encryption.**
- **Modernized insurance risk warehouse to Snowflake, implementing column-level encryption, GDPR/CCPA alignment, and full lineage documentation (Collibra), reducing audit exceptions by 45%.**
- **Prepared implementation documents for every release, worked on initial loads and data catchup process during implementations, and provided on-call support for the first few days of execution.**

Client: Ryder System, Inc

02/2020 to 01/2022

Role: Sr. Data Engineer II & OnCall Support Lead

Project: Data Modernization, PII Compliance & Governance Automation Projects

- **Developed a scalable data warehouse using Snowflake and AWS Glue, consolidating data from 30 disparate systems, resulting in a 50% reduction in report generation time and enabling self-service analytics.**
- **Developed data pipelines using Kafka Connect for real-time ingestion, Apache Iceberg ACID storage (S3), and Airflow TaskFlow API for dynamic DAGs, processing 5TB daily across multiple sources.**
- **Architected Snowflake EDW and S3 bucket for integrating data from multiple source systems which include loading nested JSON formatted data.**
- **Integrated Tableau and AWS Lambda to create a automated data refresh system, reducing manual data updates by 90% and ensuring near real-time analytics.**
- **Created Tableau Server workbooks with row-level security, ensuring data compliance and reducing unauthorized access incidents by 70%.**
- **Designed and implemented a data quality and governance framework using Python GX library and Airflow, ensuring compliance with regulations and reducing data-related audit findings by 90%.**

Client: Windhaven Insurance Company

01/2016 to 02/2020

Role: Data Engineer

Project: Data Migration, Data Quality & Regulatory Reporting Project

- **Ingested semi-structured data using snowpipes from Azure ADLS2, and used Snowflake for cloud data storage(EDW). This transition has resulted in 115% return on investment in the first year of implementation.**
- **Developed data pipelines using Azure Data Factory, Kafka, Python, and Snowflake, processing 5000 insurance claims daily, reduced processing time from 14 hrs to 3hrs.**
- **Migrated SQL Server and Teradata objects into Snowflake, by implementing SnowSQL resulting in a 40% reduction in query execution time and successful migration of 500+ tables and 1000+ stored procedures.**
- **Built efficient SSIS packages for processing fact and dimension tables with complex Transforms and type 1 and type 2 SCDs, this has decreased data ingestion timeline by 11%.**

Client: 3M

08/2015 to 01/2016

Role: Data Developer

Project: Transactional Process Data Mart (Oracle 9i/10g)

- **Prepared the test scenarios and performed unit and regression tests on various PL/SQL programs. This continuous validation decreased the problem tickets by 35%, also provided confidence to the team that the data solutions were ready for go-live with minimal risk of post-deployment issues.**
- **Identified and removed bottlenecks in the Informatica Mappings/Sessions/Worklets to effectively optimize the data flow. This led to a more efficient ETL process, which reduced the data ingestion time by 60%.**

Education

Masters : Database Management, Data security

University: Chicago State University, IL, USA

Awards

- **Spotlight Award** for outstanding leadership in Data Integration, Security measures and accomplishment of YODA project at Accenture. Year: 2023
- **Wall of Fame award** for leading critical modules (PMVMI, Fleet Tracking and Fuel Efficiency) in the large-scale data migration project at Ryder. Year: 2021
- **ACSB award** at Windhaven Insurance for successfully resolving a company-wide customer-facing commissions report and demonstrating exceptional performance. Year: 2018